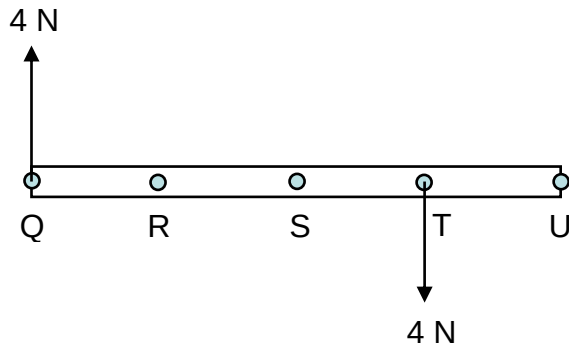


- 6 A uniform plate of weight 6 N is held horizontally by vertical forces. Forces of 4 N act at points Q and T respectively as shown in the figure below. Points Q, R, S, T and U are at equal intervals along the plate with point S located in the middle of the plate.



At what point must a third vertical force act in order to keep the plate in equilibrium?

A R

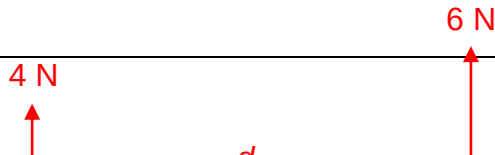
B S

C T

D U

Answer: D

The third force must be equal to the weight of the plate and must act in the upward direction.



Let d be the distance between the 4 N force at point Q and the unknown third force.

By principle of moments,

Taking moments about Q,

Sum of clockwise moments = sum of anticlockwise moments

$$6(2x) + 4(3x) = 6d$$

$$12x + 12x = 6d$$

$$d = 4x$$

Hence, the third force must act at point U.